

Course Number	0418-315	Course Title	Theory of Computation II
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Ludovit Niepel
Compulsory	No	Prerequisite	0418-211

Textbook

<i>Title:</i>	<i>Theoretical Computer Science</i>
<i>Author:</i>	<i>J.Hromkovic</i>
<i>Publisher:</i>	<i>Springer Verlag, Latest Edition</i>

Major Topics Covered in the Course

	Topics	Hours
1	Computability models: Context free grammars, parse trees, pushdown automata, languages that are not context free, algorithms for context free grammars.	6
2	Primitive recursive functions: minimization and partial recursive functions, Ackermann's function; elements of λ -calculus; RAM; Turing machine, Extensions of Turing machines, Nondeterministic Turing machines.	10
3	Church's thesis: Computing partial functions on Turing machines; halting problem, equivalence of Turing machines, RAM and partly recursive functions.	8
4	Decidability: Universal Turing machine; decidable and undecidable problems; Post's correspondence problem; undecidable problems for context-free languages, unsolvable tiling problem.	9
5	Parallel models: PRAM. CREW and CRCW models; parallel Turing machine; alternating Turing machine.	6
6	Exams.	3
	Total.	42

Grading

20%	Midterm 1
20%	Midterm 2
20%	Hmk/quiz/project
40%	Final

Course Outcomes

1. Understand pushdown automata, context-free grammars, and their normal forms.
2. Design and explain algorithms for context-free and not context free languages.
3. Understand the Church-Turing thesis and its consequences for solvability of algorithmic problems.
4. Master basic techniques used for proofs of undecidability of algorithmic problem.
5. Apply basic ideas about formal models of parallel programming and elementary techniques of designing parallel algorithms.

Relationship between Course Outcomes and Student Outcomes

Course Outcome	Student Outcomes					
	1	2	3	4	5	6
1	H					
2	L	M				
3						L
4	H					
5	H	M				M